MCQ รายวิชา 311-311 ระบบต่อมไร้ท่อและเมตาบอลิซึม

ข้อสอบมีทั้งหมด หน้า (ไม่รวมปก) จำนวน ข้อ

***ข้อสอบจริงมีจำนวน 66 ข้อ**

ชื่อ-สกุลผู้เข้าสอบ.....

เลขรหัสนักศึกษา.....

สถานที่สอบ.....

<u>สถิติการสอบ MED47 (62 คะแนน)</u>

MAX 62 MIN 22

MEAN 47.07 SD 8.80

ขอให้ผู้เข้าสอบอ่านให้ครบถ้วนและปฏิบัติตามโดยเคร่งครัด

ไม่เช่นนั้นคณะฯ จะถือว่าปฏิบัติผิดกติกาการสอบและไม่พิจารณาคะแนน

หรือให้ออกจากห้องสอบ และจะอ้างภายหลังไม่ได้ว่าไม่ทราบกติกาการสอบ

1. ห้ามเปิดอ่านหรือทำข้อสอบก่อนกรรมการคุมสอบจะประกาศว่าอนุญาตให้เปิดข้อสอบได้

2. เมื่อกรรมการคุมสอบประกาศอนุญาตให้เปิดข้อสอบได้ ให้ปฏิบัติสิ่งต่อไปนี้ก่อนเริ่มทำข้อสอบ

- นับข้อสอบว่ามีจำนวนหน้าและจำนวนข้อครบถ้วนตามที่ระบุไว้ที่ปกข้อสอบหรือไม่ หากไม่ครบถ้วนให้แจ้งกรรมการคุมสอบทันที เพื่อขอเปลี่ยนข้อสอบ หากขอเปลี่ยนภายหลังจะเสียเวลาและไม่มีการต่อเวลาให้

MCQ

- ตรวจดูเลขที่ข้อสอบบนปกข้อสอบและบนกระดาษคำตอบว่าตรงกันหรือไม่
- ใช้ปากกาเขียนชื่อ - สกุล เลขรหัสนักศึกษา ที่ปกข้อสอบ และกระดาษคาตอบ ด้วยตัวบรรจง
- อ่านคำแนะนาการใช้ดินสอดำระบายคำตอบและปฏิบัติตามให้ถูกต้อง มิฉะนั้นอาจเกิดผลเสียต่อผู้เข้าสอบเอง
- กระดาษคำตอบ ใช้ดินสอดำ 2B ระบายรหัสนักศึกษา ลงในช่องรหัสนักศึกษา ถ้าใครระบายผิด อาจไม่ได้รับการตรวจให้คะแนน เมื่อหมดเวลาแล้วไม่อนุญาตให้ย้อนกลับมาฝนรหัสต่าง ๆ อีก

Key feature หรือการสอบข้อเขียนอื่น

- การเขียนคำตอบ ใช้ปากกาสีดำหรือสีน้ำเงิน เขียนด้วยตัวอักษรที่ตัวไม่เล็กจนเกินไป ให้อ่านได้ง่าย
- ใช้ปากกาเขียนชื่อ - สกุล เลขรหัสนักศึกษา ที่ปกข้อสอบ และบนหัวข้อสอบทุกหน้า

3. ห้ามฉีกส่วนใดส่วนหนึ่งของข้อสอบออกจากชุดข้อสอบและ/หรือนำออกนอกห้องสอบ ถ้าข้อสอบหรือบางส่วนของข้อสอบชุดใดขาดหายไป จะถือว่าผู้ทำข้อสอบชุดนั้นทุจริต

4. เมื่อกรรมการคุมสอบประกาศหมดเวลาสอบ ให้วางปากกา/ดินสอทันที และนั่งรออยู่กับที่ ใครไม่หยุดทำข้อสอบจะถือว่าการสอบวิชานี้เป็นโมฆะ

5. ห้ามออกจากห้องสอบจนกว่ากรรมการคุมสอบจะเก็บข้อสอบ ตรวจนับข้อสอบและกระดาษคำตอบ เรียบร้อยแล้ว และประกาศอนุญาตให้ออกจากห้องสอบได้

1. Which of the following lipoprotein particles involved in providing energy for the cells?

A. HDL

B. IDL

C. Chylomicron

D. LDL

E. VLDL

2. Which of the following is the reason why iodine solution is indicated as sole therapy?

A. Enhanced Wolff-Chaikoff block

B. Increased vascularity of thyroid gland

C. Destroyed thyroid gland cells

D. Developed severe hypothyroidism when withdrawal

E. Developed Escape phenomenon

3. Which of the following drug should be prescribed as the adjunctive therapy for postmenopausal osteoporotic women?

A. Raloxifene + Vitamin D

B. Calcium + Vitamin D

C. Magnesium + Vitamin C

D. Zinc + Vitamin C

E. Estrogen + Vitamin D

4. A 45 years old male has been fasting for 3 days. Which of the following choices is related to this man?

A. Liver phosphoprotein phosphatase was activated resulting from insulin Liver CAMP-dependent protein kinase was inhibited resulting from glucagon

B. Affinity of liver fructose 1,6-bisphosphatase to fructose 1,6-bisphosphate is increased

C. Liver fructose 2,6-bisphosphatase was inhibited resulting from phosphoprotein phosphatase

D. Affinity of liver phosphofructokinase-1 to fructose 6-phosphate is increased

5. A 66-year-old female suffers acute L1 collapse from a simple fall. Her medical history and basic laboratory are unremarkable. Physical examination shows BMI 18 kg/m^2, mildly pale, atrophic tongue and tender point at L1. What is the most likely pathophysiologic change causing fracture of L1?

A. Iron deficiency

B. Erythroblast hyperplasia

C. Estrogen deficiency

D. Protein malnutrition

E. Folic deficiency

6. A 45 years old male with type 2 diabetes has been prescribed glibenclamide for blood glucose control. Which of the following choices is the effect of this drug on pancreatic ß cell?

A. ATP-gated K^+ channel is activated

B. Insulin granules are secreted by voltage-gated channel

C. Voltage-dependent Ca^{2+} channel is inhibited

D. Decreased intracellular concentration of Ca^{2+}

E. Pancreatic ß cell membrane depolarization

7. Which of the following lipoprotein particles increased when Apo E is abnormal?

A. IDL

B. Chylomicron

C. VLDL

D. HDL

E. LDL

8. Which of the following is the first line drug for postmenopausal women at high risk of fractures?

A. Denosumab

B. Calcitonin

C. Alendronate

D. Raloxifene

E. Teriparatide

9. A 35-year-old female presented with a 6-month history of an 8 kg body weight gain and gradual rounding of the face. Serial examinations revealed a loss of circadian rhythm of cortisol, elevated urinary free cortisol level and undetectable plasma ACTH level (<5 pg./mL). What is the most likely cause in this patient?

A. Carcinoma of bronchus

B. Adrenal adenoma

C. Pituitary tumor

D. Carcinoid tumor

E. Small cell lung cancer

10. A 10-day-old boy presents with dehydration. PE: weight 2.9 kg (birthweight 3.2 kg), dry lips, skin hyperpigmentation. Lab: Na 125, K 6.5, CO_2 15 mEq/L,
serum 17-hydroxyprogesterone 5,500 ng/dl (normal <300). Which is the gross finding of adrenal glands in this patient?

A. Atrophy both glands

B. Bilateral diffuse enlargement

C. Micronodular enlargement with pigmentation

D. One enlarged gland, one atrophic

E. Multinodular enlargement both glands

11. A 40-year-old woman presents with tiredness, anorexia and weakness for 1 year. PE: BP 86/60 mmHg, P 72/min, Skin hyperpigmentation. Lab: Na 128, K 6.5, CO2 16 mmol/L, morning serum cortisol 2.5 ug/dl (normal 5-18) and ACTH 225 pg/mL (normal 9-60). Which of the following is the most likely diagnosis?

A. Cushing's disease

B. Sheehan's syndrome

C. Waterhouse-Friderichsen syndrome

D. Addison's disease

E. Conn's syndrome

12. A 12-year-old girl was brought to a hospital with abdominal pain and eruptive xanthomas. Her fasting serum revealed milky. Which of the following choices is the most likely cause of this situation?

A. Chylomicron remnant cannot uptake into the liver

B. Chylomicron is not synthesized in the intestine

C. Cholesterol cannot efflux from peripheral cells to HDL

D. IDL cannot uptake into the liver

E. Triacylglyceride in VLDL is not hydrolyzed

13. Which of the following could be occurred if methimazole is prescribed in a 6-week pregnant patient with thyrotoxicosis?

A. Arrhythmia
B. Palpitation
C. Fetal hyperthyroidism
D. Teratogenicity
E. Liver failure

14. Which of the following lipoprotein particles involved in storing fat after eating a meal?

A. Chylomicron
B. VLDL
C. LDL
D. IDL
E. HDL

15. Which of the following hormones helps to cope with the long-term stressors?

A. Catecholamines
B. Calcitonin
C. Thyroxine
D. Sex hormones
E. Glucocorticoids

16. A 30-year-old man was diagnosed with pituitary tumor-associated acromegaly. Which of the following drugs is most likely prescribed?

A. Prednisolone
B. Goserelin
C. Octreotide
D. Mecasermin
E. Somatropin

17. What is the principal cause of metabolic acidosis of diabetes mellitus?

A. Increased triglyceride synthesis
B. Decreased gluconeogenesis
C. Increased protein synthesis
D. Decreased glycogenolysis
E. Increased lipolysis

18. Which of the following hormone levels would be compatible with hyperthyroidism due to Graves' disease?

A. High TSH but low T4 levels
B. High TSH and T4 levels
C. High T4 but low T3 levels
D. Low TSH but high T4 levels
E. Low TSH and T4 levels

19. Which of the following results is mediated by a hormone released by the posterior pituitary gland?

A. Maturation of the eggs and sperms
B. Decreased in plasma calcium levels
C. Increased in water reabsorption
D. Increased in adrenal hormone levels
E. Increased in thyroid hormone levels

20. What is the standard predictor for fragility fracture?

A. Vertebral body height 2-score
B. Bone mass density at femoral neck in T-score
C. Bone mass density at lumbar spine in Z-score
D. Age and sex
E. FRAX Thailand

21. After a car crash, the axons passing through the pituitary stalk were damaged. This damage could be expected to result in an increased releasing level of which of the following hormones?

A. Follicle-stimulating hormone

B. Growth hormone

C. Oxytocin

D. Prolactin

E. Thyroid-stimulating hormone

22. The laboratory results of the patient showed as the following:

Fasting plasma glucose	122	mg/dL
2-h 75g OGTT	227	mg/dL

What is the most likely diagnosis?

A. Addison's disease

B. Diabetic mellitus

C. Panhypopituitarism

D. Normal

E. Impaired glucose tolerance

23. A 47-year-old man has lump at neck for 2 months. PE: rubbery-firm nodule in the right thyroid lobe. The cytologic smear from fine needle aspiration reveals groups of follicular having nuclear inclusion and nuclear groove.

A. Follicular carcinoma

B. Nodular goiter

C. Subacute thyroiditis

D. Papillary carcinoma

E. Hashimoto's thyroiditis

24. A 45-year-old woman presented with a 4-year history of progressive increase in body size and a 6-week history of worsening symptoms of heart failure. Physical examination showed coarse facial features, protruding jaw, and enlarged hands and feet. On skull radiograph, destruction of the floor of the pituitary fossa with an erosion of the clinoid processes was noted.

A. Gonadotroph

B. Somatotroph

C. Corticotroph

D. Thyrotroph

E. Mammotroph

25. Gemfibrozil increases lipoprotein lipase synthesis resulting in the reduction of which substance?

A. LDL-C

B. Triglyceride

C. ApOC-III

D. HDL-C

E. PCSK9

26. Which of the following drugs decrease LDL-c by decreasing intestinal cholesterol absorption?

A. Atorvastatin

B. Alirocumab

C. Ezetimibe

D. Colesevelam

E. Niacin

27. Which of the following signs and symptoms would be found in a patient with low TSH and thyroid hormone levels?

A. Weight loss
B. Heat intolerance
C. Easily fatigued
D. Exophthalmos
E. Palpitation

28. Which of the following antidiabetic drug is the most likely cause of lactic acidosis?

A. Glipizide
B. Chlorpropamide
C. Acarbose
D. Metformin
E. pioglitazone

29. A 60-year-old man presents with severe headache PE: BP 180/100 mmHg, P 80/min. Lab: serum cortisol at 8 am 15 ug/dL (normal 6 - 23), plasma renin activity 1.2 ng/mL/hr (normal 1.9 - 3.7), serum aldosterone 44 ng/dL (normal 4.0 - 31.0), serum potassium 2.5 mmol/L. CT scan: a 2-cm mass at left adrenal gland. Which of the following is the most likely diagnosis?

A. Cortical adenoma
B. Ganglioneuroma
C. Cortical carcinoma
D. Metastatic carcinoma
E. Pheochromocytoma

30. A patient with asthma has been taking a steroidal anti-inflammatory drug for a prolonged period. Which of the following is likely to occur in hypothalamic and pituitary hormones release?

A. Low corticotropin-releasing hormone
B. Low vasopressin
C. High adrenocorticotropic hormone
D. High dopamine
E. High somatostatin

31. A 45-year-old man has a femoral neck fracture after receiving prednisolone for 6 months. What is the most likely mechanism explaining his fracture?

A. Decrease RANKL
B. Increase osteoprotogerin
C. Increase osteocalcin
D. Increase calcium absorption
E. Increase osteoblast apoptosis

32. A 20-year-old woman has xanthomas over the knees, the wrists, and elbows. Her serum LDL cholesterol is 500 mg/dL with HDL cholesterol of 40 mg/dL. She was diagnosed as having myocardial infarction recently. What is the mechanism of her dyslipidemia?

A. Decreased chylomicron synthesis
B. Increased hepatic lipid synthesis
C. Decreased lipid excretion
D. Increased hormone sensitive lipase activity
E. Decreased receptor-mediated endocytosis of LDL

33. A 37-year-old woman presented with palpitations, shortness of breath and headache. On physical examination, her heart rate was 110 bpm and the blood pressure was 170/110 mmHg. The laboratory results of plasma sodium and potassium showed as the following. What is the most likely diagnosis?

A. Secondary adrenal insufficiency

B. Secondary hyperaldosteronism

C. Primary adrenal insufficiency

D. Primary hyperaldosteronism

E. Congenital adrenal hyperplasia

34. Which of the following is a main mechanism of glyburide glibenclamide lower blood glucose levels?

A. Increase insulin synthesis

B. Promote glucose uptake by muscle, liver, and adipose tissue

C. Enhance renal excretion of glucose

D. Decrease insulin resistance by lowering body weight

E. Release insulin from the pancreas

35. Which of the following molecules is decreased in postmenopausal women and led to osteoporosis?

A. Alkaline phosphatase

B. RUNX2 (Runt-related transcription factor 2)

C. RANK (Receptor activator of NF-kB)

D. OPG (Osteoprotegerin)

E. RANKL (Receptor activator of NF-kB ligand)

36. Which of the following mechanisms involved in glucose homeostasis during fed state?

A. Increased glucose uptake in brain tissue

B. Increased hepatic glucose production

C. Increased hepatic glycogenolysis

D. Increased urine excretion of glucose

E. Increased glucose uptake in muscle tissue

37. Which of the following conditions increases plasma insulin levels?

A. Hypoglycemia

B. After intravenous administration of somatostatin

C. Following vigorous exercise

D. Following a carbohydrate-rich meal

E. During a surgical procedure

38. The combination of simvastatin and which drug may increase risk of statin-associated myopathy?

A. Ezetimibe

B. Ketoconazole

C. Codeine

D. Cholestyramine

E. Rifampicin

39. Which of the following laboratory results indicates the highest risk of coronary heart disease?

A. HDL-C = 30 mg/dL and Total cholesterol = 260 mg/dL

B. HDL-c = 60 mg/dL and Total cholesterol = 190 mg/dL

C. HDL-c = 55 mg/dL and Total cholesterol = 170 mg/dL

D. HDL-C = 35 mg/dL and Total cholesterol = 220 mg/dL

E. HDL-C = 40 mg/dL and Total cholesterol = 180 mg/dL

40. A 30-yr-old female presented with significant weight gain, dry skin, hair loss, milky discharge and goiter. The laboratory test showed FT4 0.2 ng/dL (0.7-1.4), TSH > 100 mlU/L (0.5-5), Prolactin 110 mcg/L (0-25). What is the mechanism of hyperprolactinemia?

A. Low clearance of prolactin

B. Undetected prolactinoma

C. Low FT4

D. High TSH

E. High TRH

41. A 45-year-old man with type 2 diabetes presents with unconsciousness and plasma glucose level of 41 mg/dL. Which of the following drugs is most likely to cause hypoglycemia?

A. Exenatide

B. Pramlintide

C. Pioglitazone

D. Voglibose

E. Glibenclamide

42. What is the diagnosis in a patient who has hyperglycemia from total pancreatectomy and requires insulin injection?

A. Type 1 diabetes

B. Diabetes of undetermined cause

C. Type 2 diabetes

D. Other specific type of diabetes

E. Insulin dependent diabetes mellitus

43. Which of the following mechanisms involved in normalizing plasma glucose in subjects with insulin resistance?

A. Decreased appetite

B. Decreased intestinal glucose reabsorption

C. Increased urinary glucose excretion

D. Increased insulin secretion

E. Increased glucose uptake in brain tissue

44. A 40-yr-old female presented with Cushingoid appearance. After taking dexamethasone 1 mg at 11 pm, the result is very high morning cortisol in the following day. What is the mechanism of this result?

A. Loss of circadian rhythm

B. Increase production of cortisol

C. Loss of negative feedback

D. Interference of cortisol assay by dexamethasone

E. Increase cortisol binding protein

45. Which of the followings acts as a tropic hormone from anterior pituitary gland?

A. Prolactin-inhibiting hormone

B. Thyrotropin-releasing hormone

C. Luteinizing hormone

D. Cortisol

E. Corticotropin-releasing hormone

46. Which enzyme is important for signal transduction of growth hormone receptors?

A. Phospholipase A2

B. Janus kinase/signal transducer and activator of transcription

C. Adenylyl cyclase

D. Phospholipase C

E. Guanylyl cyclase

47. A 20-year-old female patient was diagnosed with Addison's disease. She is treated with hydrocortisone but still has hyponatremia. Which of the following medications should be added to this patient?

A. Prednisolone

B. Synthetic ACTH

C. Abiraterone

D. Fludrocortisone

E. Dexamethasone

48. The 50-year-old woman presented with headache, palpitation, sweating, pallor, tremor and abdominal discomfort. Her clinical features suggested pheochromocytoma. Which of the following laboratory tests should be further investigation?

A. Plasma ACTH

B. Plasma aldosterone

C. Serum cortisol

D. Urine vanillylmandelic acid

E. 24-h Urine free cortisol

49. A 23-year-old man presented with nausea and vomiting. Laboratory results revealed plasma fasting glucose 420 mg/dL (N 70 - 99 mg/dL), plasma HCO3-6 mmol/L (N 22 - 26 mmol/L), and urine ketone with 4+. What is the most likely diagnosis?

A. Impaired glucose tolerance

B. Hyperosmolar hyperglycemic syndrome

C. Diabetes insipidus

D. Non-ketotic hyperglycemia

E. Diabetic ketoacidosis

50. A female patient with Cushing's syndrome complains of hirsutism and acne after receiving a drug for inhibiting cortisol synthesis. Which of the followings is she most likely taken?

A. Mifepristone

B. Abiraterone

C. Ketoconazole

D. Mitotane

E. Metyrapone

51. Which of the following choices is a receptor responsible for paracrine hormone induction of osteoclastogenesis?

A. RANK (Receptor activator of NF-kB)

B. OPG (Osteoprotegerin)

C. RANKL (Receptor activator of NF-kB ligand)

D. Alkaline phosphatase

E. RUNX2 (Runt-related transcription factor 2)

52. A 58-year-old man had a thyroid cancer removal. Six months later, he had a regular medical check-up. The doctor mentioned that he had an abnormal calcium level (2.0 1.5 mmol/L). Which of the following signs/symptoms might be observed?

A. Dehydration

B. Tetany

C. Osteoporosis

D. Kidney stones

E. Constipation

53. A 3-year-old girl presented with bone pain, abnormal bone shape and slow growth rate. The doctor informed parents that she had hypophosphatemic condition due to a massive loss of phosphate in the urine. Which of the followings might relate to her signs and symptoms?

A. Increased 24(OH)ase expression in the kidneys

B. Decreased plasma FGF-23 levels

C. Increased lla Na+/Pi cotransporter expression in the kidneys

D. Increased llb Na+/Pi cotransporter expression in the intestine

E. Decreased plasma calcitonin levels

54. A 50-year-old man comes for check-up. Laboratory findings: serum calcium 12 mg/dL (normal 8.0-9.5), phosphate 2.8 mg/dL (normal 3.5-5.5), parathyroid hormone 70 ng/L (normal 10-65). Which of the following condition/disease the patient is most likely to have?

A. Nodular enlargement of 4 parathyroid glands

B. Medullary carcinoma of thyroid

C. Calcification of parathyroid gland

D. Nodule in one parathyroid gland

E. chronic kidney disease

55. Which of the following processes would be most affected in lecithin cholesterol acyl transferase (LCAT) deficiency?

A. Binding of LDL to its receptor

B. Conversion of discoidal HDL to HDL3

C. Conversion of chylomicron-to-chylomicron remnant

D. Uptake of HDL into the liver

E. Transfer of cholesterol from LDL to peripheral tissues

56. Which of the following glucose transporters plays a central role in glucose tolerance?

A. GLUT 6

B. SGLT 2

C. SGLT 1

D. GLUT 2

E. GLUT 4

57. After a car crash, the axons passing through the pituitary stalk were damaged. This damage could be expected to result in an increased releasing level of which of the following hormones?

A. Prolactin

B. Oxytocin

C. Growth hormone

D. Thyroid-stimulating hormone

E. Follicle-stimulating hormone

58. Which of the following is a mechanism of aldosterone to trigger cellular response?

A. Activation of adenylyl cyclase

B. Forming hormone-receptor complex in cytoplasm

C. Activation of phospholipase C

D. Binding to G-protein coupled receptors

E. Autophosphorylation of the receptors

59. Which of the following is the inhibition site of methimazole?

A. Iodine organification

B. Thyroid hormone release

C. Peripheral conversion

D. Iodine reduction

E. Iodide trapping

60. The laboratory results of the patient showed as the following:

Total cholesterol	300	mg/dL
Triglyceride	220	mg/dL
HDL-c	50	mg/dL

Which of the following LDL-c results should be found in this patient?

A. 226 mg/dL

B. 30 mg/dL

C. 80 mg/dL

D. 126 mg/dL

E. 206 mg/dL

61. Which of the followings is the mechanism of thiazide diuretics on calcium reabsorption in the kidneys?

A. Increased Na+-Cl- cotransporter activity

B. Decreased Na+-K+-2Cl- cotransporter activity

C. Decreased TRPV5 activity

D. Increased NCX1 activity

E. Increased Ila Na+/Pi cotransporter expression

62. Which of the following drugs is most likely prescribed to decrease testosterone production in patient with prostate cancer?

A. Bromocriptine

B. Mifepristone

C. Lanreotide

D. Urofollitropin

E. Leuprolide

63. A 20-year-old woman presents with neck pain for 2 days. PE: P 110/min, mild enlargement of thyroid gland with marked tenderness. Lab: free T4 1.90 ng/dL (normal 0.7-1.7), TSH 0.05 mIU/L (normal 0.25-4.00). Which of the following is the pathological characteristic of this thyroid disease?

A. caseous necrosis

B. Hurthle cell

C. multinucleated giant cells engulfing colloid

D. lymphocyte follicles

E. bacterial organism in follicles

*หมายเหตุ ข้อสอบจริงมี 66 ข้อ แต่มีข้อซ้ำกับข้อฟรีทำให้คะแนนเต็มเหลือแค่ 62 คะแนน

VICHAKAN MED 47